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A. Limitations and Future Work

Going beyond Bongard Problems. BPs fundamentally require solvers to articulate answers in natural language, making them a valuable testbed for assessing the reasoning capabilities of MLLMs. However, to comprehensively explore the challenges posed by the AVR domain, it is crucial to consider a broader range of problems.

For instance, VCog-Bench (Cao et al., 2024) is a benchmark designed to evaluate the reasoning capabilities of MLLMs across 3 datasets: 560 problem instances from RAVEN (Zhang et al., 2019), 309 from CVR (Zerroug et al., 2022), and 480 from MaRs-VQA. These datasets present multi-class classification tasks, offering between 4 to 8 options per problem. While the classification setting in VCog-Bench differs from our focus on natural language generation, both studies echo a shared conclusion – MLLMs struggle in complex, multi-image reasoning tasks. We argue that generative problem formulations, such as those used in our study, pose a more substantial challenge than discriminative tasks, in which the solution may be induced from correlations or by making educated guesses. Further advances in abstract reasoning may require the development of new AVR benchmarks with generative evaluation settings.

A compelling example of a generative problem formulation is the Abstraction and Reasoning Corpus (ARC) (Chollet, 2019), in which each instance involves transforming a source grid into a target grid based on an induced transformation rule. Each instance is accompanied by a few demonstrations to guide the solver. (Mitchell et al., 2023) explored multi-modal reasoning of GPT-4V on ConceptARC (Odouard & Mitchell, 2022), a variant of ARC categorizing tasks into distinct types. The study employed 3 prompting strategies: presenting all demonstrations in a single image, using separate images for each source and target grid pair, and separating each grid pair into distinct images. These settings are related to the Direct, Contrastive, and Descriptive strategies from our study, resp. The model performed best with the last approach, which aligns with the leading performance of the Descriptive strategy in our paper. Their study revealed that ARC tasks pose a significant challenge for MLLMs, aligning with our results. Similar to our findings, GPT-4V evaluation on ConceptARC demonstrated that generative problem formulations pose a significant challenge for contemporary MLLMs.

Fine-grained analysis of MLLM perception. Related studies emphasize the importance of evaluating fine-grained aspects of model performance in visual reasoning tasks. Notably, (Biscione et al., 2024) propose the MindSet: Vision toolbox, which categorizes tasks into three main domains: low- and mid-level vision, visual illusions, and shape and object recognition. This benchmark is specifically designed to test models on 30 psychological findings inspired by human visual perception, providing a framework for understanding similarities and differences in human and machine vision. Preliminary evaluations using ResNet-152 and GPT-4 on selected tasks revealed notable differences in perception between humans and machines. Applying MLLMs and the reasoning strategies proposed in our work to the MindSet: Vision toolbox opens a promising direction for future research, which could offer deeper insights into the perceptual capabilities of MLLMs.

Incorporating proposed strategies to enhance abstract reasoning abilities. (Galatzer-Levy et al., 2024) compared the cognitive abilities of MLLMs to humans using the Wechsler Adult Intelligence Scale (WAIS-IV) (Wechsler, 2008). Their findings reveal that while MLLMs excel in tasks related to verbal comprehension and working memory, they significantly underperform in perceptual reasoning tasks. The evaluation setting used in this study involved presenting models with an image of an abstract reasoning matrix alongside a text prompt describing the task, closely aligning with the Direct strategy employed in our work. However, as discussed in Section 5, the Direct strategy poses notable challenges for MLLMs. Our experiments show that models consistently achieve better performance with alternative approaches, such as the Descriptive strategy. This highlights the importance of selecting appropriate strategies when evaluating MLLMs on abstract reasoning tasks. We believe that the diverse suite of strategies proposed in our work can be extended to other studies in abstract reasoning to fully capitalize on MLLM reasoning capabilities.

Cross-domain analysis of MLLM perception. A possible hypothesis for the subpar performance of MLLMs on AVR tasks involving synthetic datasets, such as VCog-Bench or ARC, is the limited representation of synthetic images in their training data. This assumption is supported by the observed performance gap between synthetic BPs and real-world image BPs, such as those in Bongard HOI and Bongard-OpenWorld, which might suggest that MLLMs perform better at abstract reasoning with real-world images. However, our experiments with Bongard-RWR challenge this notion. Despite using real-world images, Bongard-RWR demonstrates that MLLMs still struggle with abstract reasoning, indicating that the performance gap cannot be solely attributed to differences in data domains. Instead, this suggests more fundamental challenges in visual reasoning. Future work could extend this research line by leveraging datasets that include both synthetic and real-world images, such as Raven’s Progressive Matrices (Zhang et al., 2019; Teney et al., 2020) or Visual Analogy

Prompt 1: Text used as a brief introduction in human testing.

The presented problems represent a type of logic puzzle. Each problem consists of two
 ↳ sides separated by a vertical line. Each side contains six images. The task is to
 ↳ find a characteristic that applies to all the images on the left side but does not
 ↳ apply to those on the right side.

Some problems may be less obvious and require a broader perspective or focus on details.
 ↳ Simply comparing the general content of the images might not be enough. Answers may
 ↳ repeat.

Problems (Hill et al., 2019; Bitton et al., 2023). Contrasting MLLM performance on such dataset pairs may provide valuable insights into whether their limitations are rooted in data domain or in broader domain-free reasoning challenges.

B. Human Performance on Bongard-RWR

Foundation of the study. Our tests on MLLMs using the Bongard-RWR dataset revealed their poor performance in solving synthetic concepts depicted in real-world images. However, the difficulty and reliability of this new dataset remains an open question. To address this issue, we decided to assess human capabilities in solving these problems.

Methodology. We compiled all Bongard-RWR problems into a single document, including a brief introduction that explains what BPs are (see Prompt 1), along with a few detailed examples. The examples included one problem from the original BPs (#133), one from Bongard-OpenWorld, and an additional BP (#336) manually translated to the real-world domain. Bongard-RWR problem instances were positioned randomly in the document and were posed in an open-ended manner, allowing participants to provide any response they deemed appropriate.

Participants in our human evaluation predominantly belonged to the academic community, including Master students and (occasionally) faculty members and PhD students, primarily due to accessibility. This demographic was selected based on the ease of reaching and engaging with individuals who are readily available in academic settings.

All answers were collected using an online form, ensuring a streamlined and efficient process for submission. Each participant was allowed to make only a single submission, to maintain the integrity and reliability of the data. In addition, the form contained a few more questions to gather basic statistics on our new dataset and the quality of submissions:

1. How would you assess the readability of the images included in the problems? (Scale 1 – 10)
2. How would you assess the difficulty of the tasks you received? (Scale 1 – 10)
3. What is your level of education? (Primary, Secondary, Higher, I prefer not to say)
4. How much time did you spend solving the tasks? (Less than 30 minutes, From 30 minutes to an hour, From one to two hours, More than two hours)

Answers evaluation. In contrast to the evaluation of MLLM solutions, human responses were evaluated entirely manually. Initially, two humans reviewed the complete set of answers independently, achieving a 94.5% agreement on the correctness of the responses. The discrepancies were then discussed and a consensus was reached leading to a single, unified evaluation.

Respondent overview. Overall, we successfully collected 30 responses, with 26.7% of participants having secondary education and 73.3% having higher education. Although the sample of respondents was small, we observed no significant discrepancies between the results of individuals with higher education and those with secondary education, both in the responses to additional questions (Fig. 6) and in the problem-solving results (Fig. 8).

Findings from respondent responses. In Fig. 8, we present the distribution of responses to the Bongard-RWR dataset. Every problem was solved by at least one respondent, confirming the solvability of the dataset. The results are consistent across respondents (see Fig. 7), with the number of solved problems ranging from 23 to 59. Moreover, the findings